Paired Activity: Disaster Response and Recovery

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Background

A major hurricane causes significant flooding across a coastal region affecting three counties. Six months after the disaster, a research team wants to study household recovery, specifically which households have recovered financially and physically, and what factors predict faster recovery. The affected area includes approximately 45,000 households. Many have temporarily relocated. Some neighborhoods were completely destroyed while others had only minor damage.

Part A — Sampling Method

1. What challenges does this disaster context create for sampling that is representative of the population?

2. Would systematic sampling work here (e.g., sampling every 75th household from an address list)? Why or why not?

3. Some researchers propose sampling only households that are *still in the area* and have not relocated. What kind of bias does this introduce? What type of households are likely missing from your sample? Can you propose any alternatives?

Part B — Study Design

The team wants to understand whether households that received **case management support** (a social service connecting families to resources) recovered faster than those who did not.

1. Is an RCT ethical and feasible in this post-disaster setting? Why or why not?

2. Design a **cohort study** for this question. Mention who will be in your cohort, what is the **exposure**, and the **outcome**? When and how often might you measure and how long do you follow up?

3. What is the biggest confounding threat to a cohort design in this setting?

Part C — Geographic Considerations

(Think about this together — we will return to it next class)

Households that flooded on the same street shared the same flood depth, the same proximity to emergency services, and the same neighborhood social networks.

Discuss the following with your partner:

- If you randomly sample 200 households from the flooded area, are those 200 households giving you 200 independent pieces of information — or something less?

- How might the spatial pattern of damage (some blocks completely destroyed, others barely touched) affect how you design your sample?

- Does where a household sits relative to a flood boundary tell us something important that gets lost when we just record "flooded = yes/no"?

No right or wrong answers here — just think out loud. We will explore these ideas in our next class.

Paired Activity: Wildfire Management

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Background

Wildlife managers want to estimate the population size and habitat use of a declining grassland bird species across a large, protected reserve of 2,000 square kilometers. The reserve contains a mix of habitat types: dense grassland, shrubland, wetland edges, and degraded areas near the reserve boundary. Field surveys are expensive — each survey point requires a trained observer spending 20 minutes recording birds. The team can conduct 150 survey points per season.

Part A — Sampling Method

1. The team considers placing survey points using simple random sampling across the reserve. What is the risk of this approach given the different habitat types present?

2. How would stratified sampling by habitat type improve your estimate? What information do you need *before* you can implement stratified sampling?

3. Could a systematic sample using a regular grid of survey points work here**?** List at least two advantages and two potential weaknesses.

Part B — Study Design

Managers want to know whether an invasive grass removal program implemented in one section of the reserve has increased bird density compared to untreated sections.

1. Sketch a simple study design. What is the **treatment**? What is the **control**? What do you measure and when?

2. Managers were not able to randomly assign which sections received treatment — they chose the most degraded sections first. What study design problem does this create?

3. What would a **before-and-after with control group** design look like here? Draw a simple timeline showing when you would collect data. Is this stronger than a simple comparison of treated vs. untreated after treatment?

Part C — Geographic Considerations

(Think about this together — we will return to it next class). Discuss the following with your partner:

- Birds don't respect the boundaries of your sampling strata or treatment/control sections. A bird detected at a survey point in the treated section may have flown in from an untreated section. Does this matter for your study conclusions?

- Survey points close to each other are more likely to detect the *same individual bird* than points far apart. What does this mean for how much information your 150 survey points are actually giving you?

- If you find higher bird density near the center of the reserve than the edges, is that because of habitat quality — or could something else explain it?

No right or wrong answers here — just think out loud. We will explore these ideas in our next class.

Paired Activity: Environment Health and Pollution

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Background

Researchers want to study whether long-term exposure to air pollution from industrial facilities is associated with higher rates of childhood asthma in a mid-sized metropolitan area. The metro area has 12 industrial facilities distributed unevenly across the region. Demographic data suggest that lower-income and minority communities are more likely to live near these facilities. Health records are available at the zip code level. School health records contain asthma diagnosis data for approximately 40,000 children.

Part A — Sampling Method

1. The researchers have access to health records for *all* 40,000 children in the metro area. Does having the full population mean they no longer need to think carefully about sampling or statistical issues? Explain your answer.

2. If budget limits the team to a sample of 2,000 children, compare **simple random sampling** with **stratified sampling by proximity to industrial sites**. What might be the advantages and risk of each method?

3. What information would you want to know about all 40,000 children before designing your sampling strategy?

Part B — Study Design

1. Is this research question suited to an **RCT**? Explain clearly why or why not.

2. Design a **cohort study** to answer this question. Who is in your cohort? What counts as **exposed**? What is the **outcome**? What **confounders** might you measure?

3. A colleague suggests a **cross-sectional design** instead — measuring both pollution exposure and asthma rates at one point in time using zip code data. What can this tell you? What can it *not* tell you?

Part C — Geographic Considerations

(Think about this together — we will return to it next class)

You measure pollution exposure using zip codes, specifically whether a child's zip code contains an industrial facility.

Discuss the following with your partner:

- A child living one block from a factory but on the "wrong side" of a zip code boundary is classified as unexposed. A child 3 miles from a factory inside the same zip code is classified as exposed. What is wrong with this approach?

- Children who live near each other breathe the same air and have similar environments. Are these observations independent? Why is this important?

- Lower-income families are more likely to live near facilities. How does this make it very difficult to separate the effect of pollution from the effect of poverty?

No right or wrong answers here — just think out loud. We will explore these ideas in our next class.

Paired Activity: Energy Access and Burden

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Background

A state energy office wants to understand household energy burden (the percentage of household income spent on energy bills) to identify which communities should be prioritized for weatherization subsidies and rooftop solar programs. The state has a diverse geography consisting of dense urban centers, mid-sized cities, rural agricultural areas, and remote tribal lands. Administrative data on energy bills is available for utility customers but does not cover households using propane or wood fuel, which are common in rural areas.

Part A — Sampling Method

1. The administrative utility dataset covers about 80% households in the state. What types of households are likely in the missing 20%, and what kind of **sampling bias** does this create?

2. Design a sampling strategy that would give you reliable estimates for *each* of the four geographic area types: urban, mid-sized city, rural, and tribal lands. Name the method, explain how you would implement it, and justify your choice.

3. If you use **cluster sampling** by utility district, what might be the main concern?

Part B — Study Design

The energy office ran a **pilot** weatherization program in 8 randomly selected counties last year and wants to evaluate whether it reduced household energy burden. Eight similar counties that did not receive the program serve as a **comparison** group.

1. Is this an RCT? Or where does it fall short? Think about whether these is **random assignment, control group, baseline measurement, and blinding**.

2. What information would you collect *before* the program started (baseline) and *after* to evaluate impact? Why is baseline data critical?

3. The 8 treatment counties were randomly selected but happen to be geographically clustered in the colder northern part of the state. Does this matter generalizability?

Part C — Geographic Considerations

(Think about this together — we will return to it next class)

Energy burden is shaped heavily by where you live — climate zone, housing stock, distance from energy infrastructure, and local utility pricing.

Discuss the following with your partner:

- Two households with identical incomes may have very different energy burdens simply because one is in a cold northern county and one is in a mild southern county. How does this complicate comparisons across the state?

- Your **sampling frame** misses propane and wood fuel users. Where do these households tend to be located geographically? What does their absence mean for your policy conclusions?

- If you find rural households have higher energy burden than urban households, is that because of income, climate, housing type, or all three simultaneously? How would you even begin to separate these?

No right or wrong answers here — just think out loud. We will explore these ideas in our next clPaired Activity: Transportation and Accessibility

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Background

A metropolitan planning organization (MPO) wants to assess transportation accessibility for older adults (65+) — specifically whether they can reach essential services such as medical appointments, grocery stores, and pharmacies using public transit or non-motorized travel. The metro area has well-developed transit in the urban core but sparse coverage in outer suburbs. Approximately 180,000 residents are 65+, distributed unevenly across the metro area.

Part A — Sampling Method

1. The MPO proposes recruiting interview participants at senior centers and medical clinics. What **sampling bias** does this introduce? Who is likely *missing* from this sample?

2. Compare **stratified sampling** by residential zone (urban core / inner suburb / outer suburb) with **simple random sampling** from a voter registration list. What might an advantage and limitation of each approach. Who might be missed?

3. The MPO can afford 300 interviews. Older adults are not evenly distributed — they are concentrated in some inner suburbs and newer outer suburbs. How would you decide how many interviews to conduct in each zone? What principle guides this decision?

Part B — Study Design

The MPO introduced an on-demand app-based shuttle service in three outer suburban areas as a pilot. They want to know if it improved accessibility for older adults.

1. How might you evaluate this intervention? What outcomes would you measure? What is your comparison group?

2. The three pilot areas were chosen because they had the worst existing transit access in the metro area. How does this selection method affect your ability to compare them to non-pilot areas?

3. Older adults who are more mobile and comfortable with technology may be more likely to use the new service. How does this **self-selection/volunteer bias** affect how you interpret the results?

Part C — Geographic Considerations

(Think about this together — we will return to it next class)

You measure "transit access" using a simple rule: does the person live within 400 meters of a bus stop?

Discuss the following with your partner:

- Two people both live 390 meters from a bus stop. One has safe, flat, well-lit sidewalks. The other has no sidewalks, a busy road to cross, and limited mobility. The 400-meter rule classifies both as having access. What is the problem here?

- Older adults in the outer suburbs are spread across large areas. If you survey 50 of them in one outer suburb, are they giving you 50 independent perspectives — or are they all experiencing the same transportation environment in similar ways?

- The pilot micro-transit zone has a specific geographic boundary. A resident just outside gets no service; a resident just inside does. Is that boundary meaningful in terms of how people actually move and live?

No right or wrong answers here — just think out loud. We will explore these ideas in our next class.

Paired Activity: Climate & Heat Adaptation

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Background

A city health department wants to understand how residents are coping with increasingly frequent extreme heat events. They want to know whether people are using cooling centers, modifying their behavior, or making home adaptations such as installing fans or improving insulation. The city has a population of 400,000 spread across dense urban neighborhoods, mid-density suburbs, and rural fringe areas. The budget allows for 600 survey interviews.

Part A — Sampling Method

The research team is debating between three approaches:

- Simple random sampling from the city's resident registry
- Stratified sampling by neighborhood type (urban core / suburban / rural fringe)
- Cluster sampling by city block

Questions:

1. What are the strengths and weaknesses of each sampling approach for this specific study?

2. Which method would you recommend and why?

3. Is 600 interviews enough? What factors should guide that decision?

Part B — Study Design

The health department also wants to know whether people who received a city-mailed "heat preparedness kit" last summer actually changed their behavior compared to those who did not receive one.

1. What study design would best answer this question? Explain your reasoning.

2. Could this be structured as an RCT? What would that require, and what challenges would arise?

3. If an RCT is not feasible, what is your next best option and what are its limitations?

Part C — Geographic Considerations

(Think about this together — we will return to it next class)

You decide to use stratified sampling, dividing the city using zip codes as your neighborhood type boundaries.

Discuss the following with your partner and jot down your initial thoughts:

- Does it matter which geographic boundaries you use to define "neighborhood type"? What if you had used census tracts instead of zip codes?

- Two neighbors live on opposite sides of a zip code line and are classified into different strata. Does that make sense?

- People in the same neighborhood share parks, the same urban heat island effect, and similar building types. Does that create any problem for how you interpret your results?